

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Quadratic Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING QUADRATIC EQUATIONS · CH2 · L10

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The Map

This lesson collapses Solving Quadratic Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Factorise then solve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the quadratic formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Complete the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Use graph roots

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 4 website questions, 4 checked solutions, 3 local PDFs read.

01 Factorise then solve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use the quadratic formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Complete the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 21x + 20 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q9

Solve quadratic equation

$$x^2 - 21x + 20 = 0$$

Factorise

$$(x - 20)(x - 1) = 0$$

Finish: $x = 20$ or $x = 1$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

| Solve $x^2 - 7x + 10 = 0$.

04 Use graph roots

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5x^2 + 9x - 17 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1H Q15

Solve quadratic equation

$$5x^2 + 9x - 17 = 0$$

Use quadratic formula

$$x = \frac{-9 \pm \sqrt{9^2 - 4(5)(-17)}}{2(5)}$$

Finish: $x = 1.15$ or $x = -2.95$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Factorise then solve
"work out", "show", "hence", a familiar exam phrase	Use the quadratic formula
"work out", "show", "hence", a familiar exam phrase	Complete the square
"work out", "show", "hence", a familiar exam phrase	Use graph roots
"work out", "show", "hence", a familiar exam phrase	Solve quadratic equation
"work out", "show", "hence", a familiar exam phrase	Factorise

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.